AWS Serverless EC2 Start-Stop Scheduler using Terraform

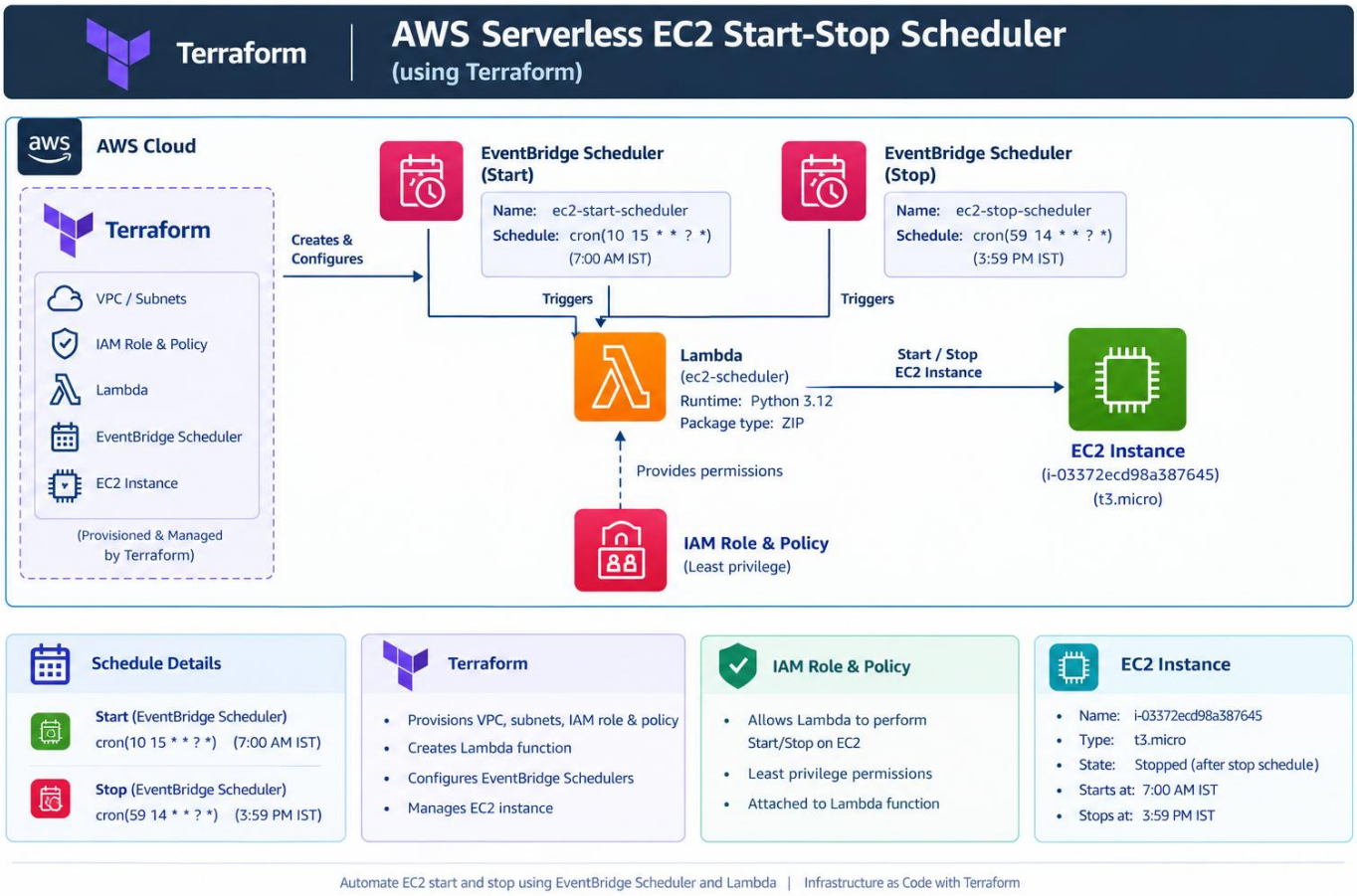
Prepared By: Pooja Daingade

Project Overview

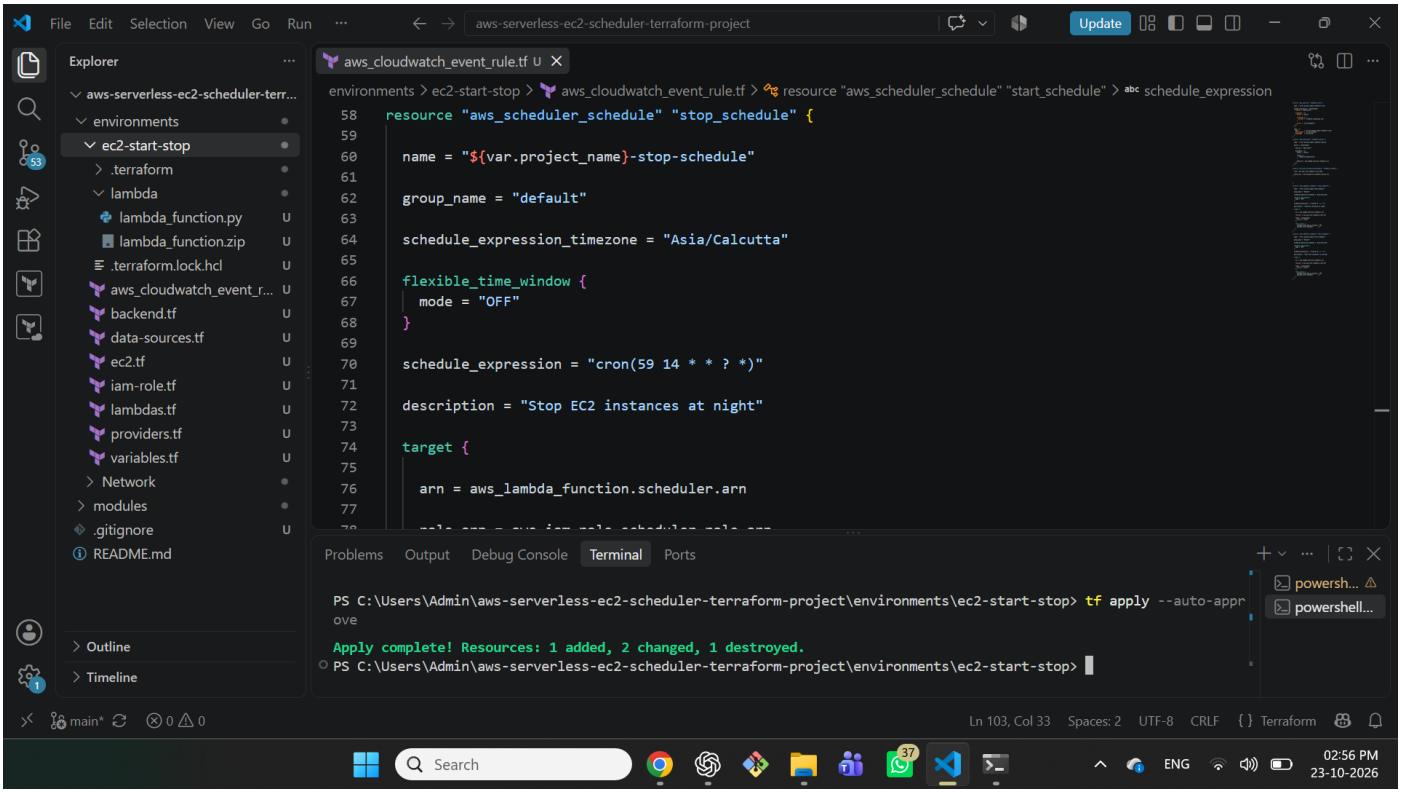
The AWS Serverless EC2 Start-Stop Scheduler using Terraform automates EC2 instance management using AWS Lambda and Amazon EventBridge Scheduler. Terraform is used to provision and manage the required AWS resources.

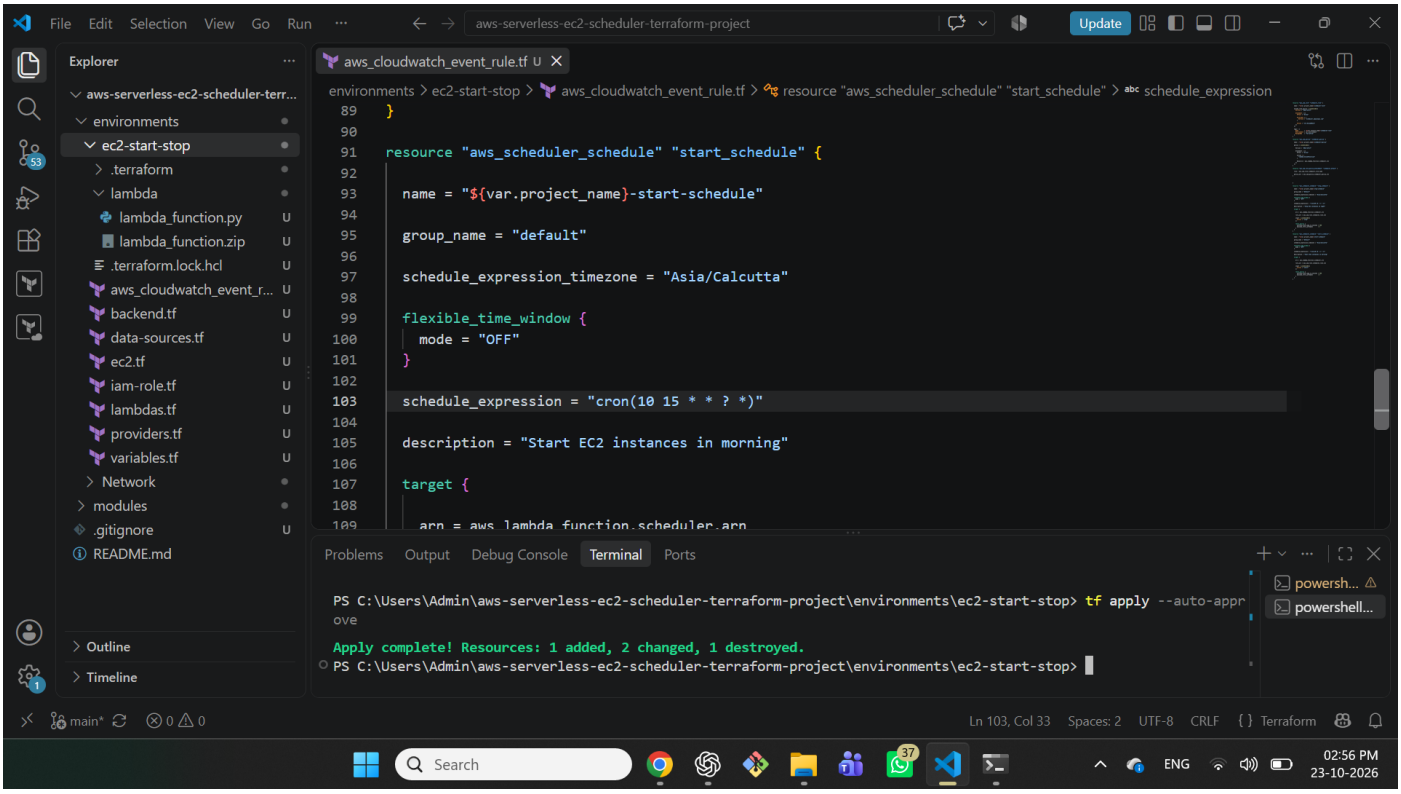
EventBridge Scheduler triggers the Lambda function at predefined times to start and stop EC2 instances automatically. IAM roles and permissions provide controlled access to EC2 operations. This solution reduces manual intervention, improves operational efficiency, and helps optimize AWS compute costs by running instances only when required.

Architecture Diagram

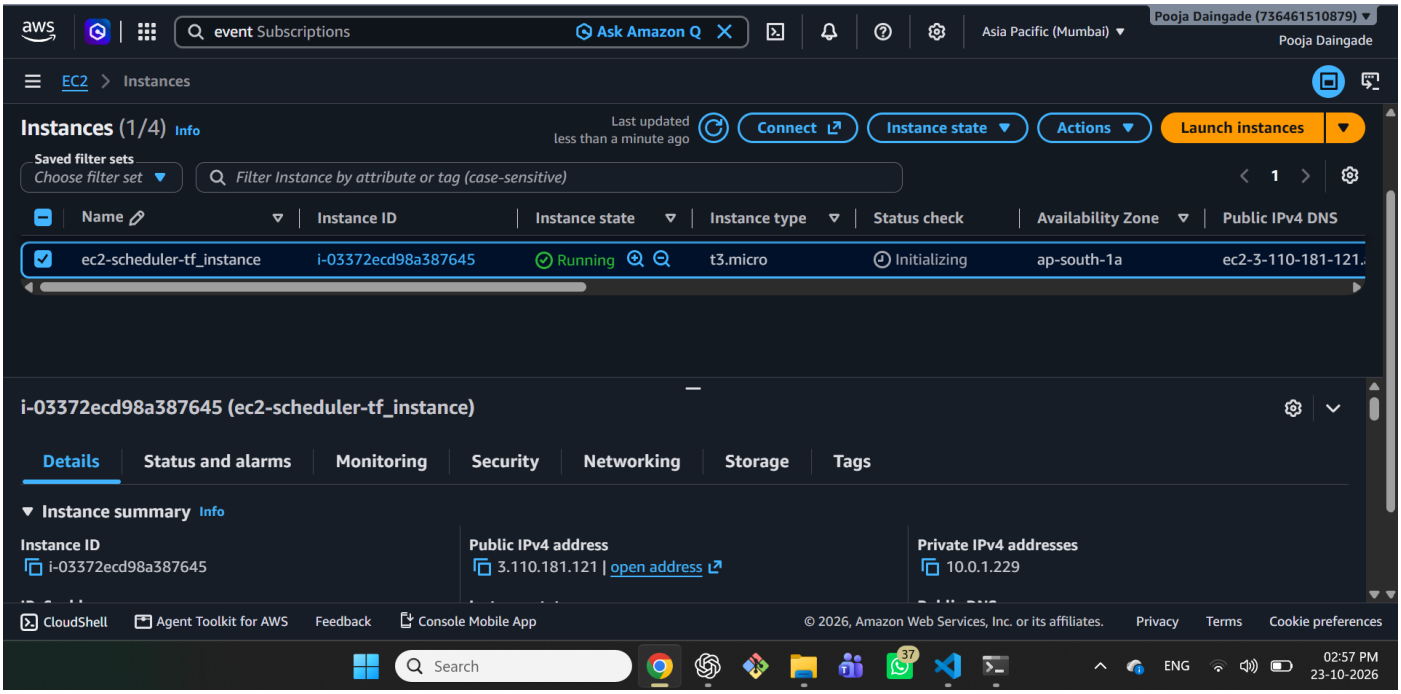


1. Terraform is used to deploy and manage the AWS Serverless EC2 Start-Stop Scheduler infrastructure. The configuration automates EC2 instance start and stop operations using AWS Lambda and Amazon EventBridge Scheduler.

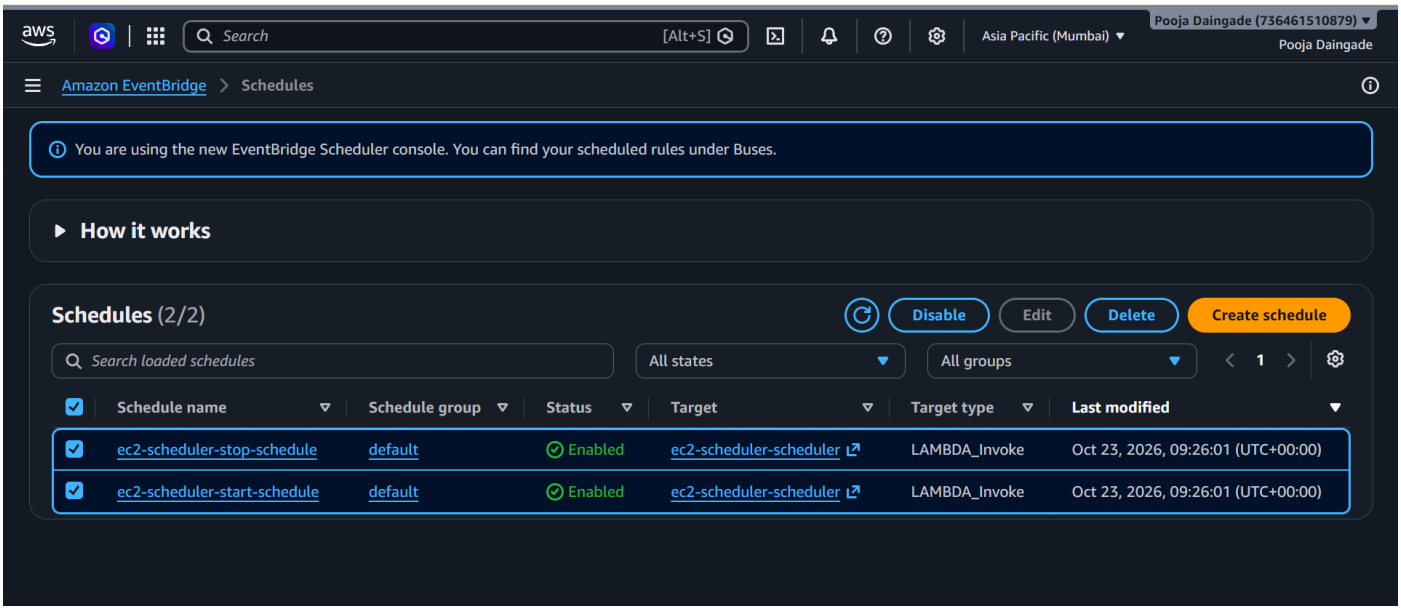




2. After executing terraform apply, the EC2 instance was successfully provisioned and entered the **Running** state.

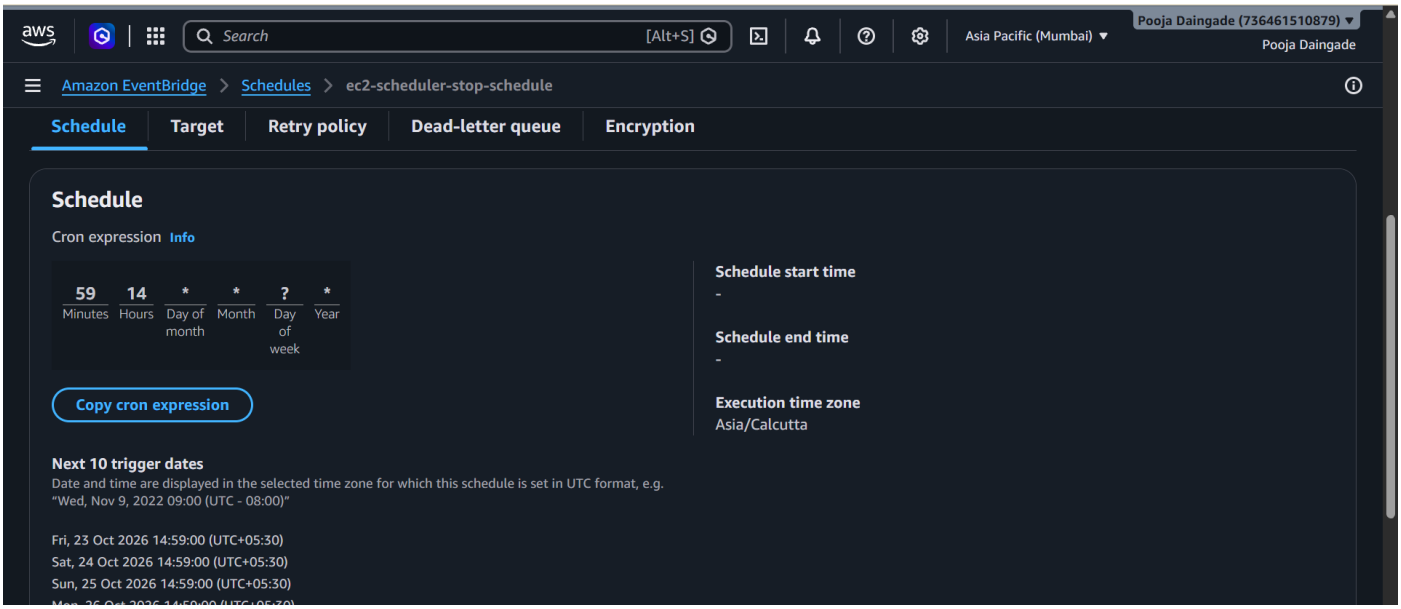


3. Both schedules are successfully created and enabled to automate EC2 instance start and stop operations.
   Start Schedule: ec2-scheduler-start-schedule
   Stop Schedule: ec2-scheduler-stop-schedule



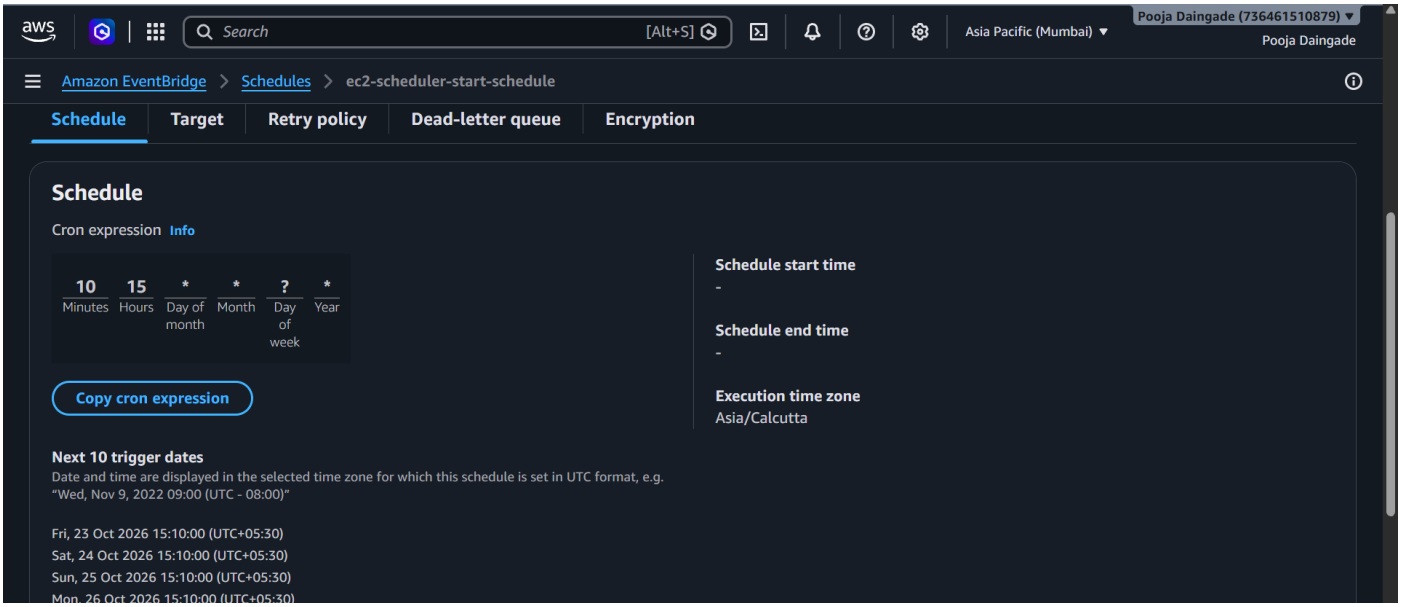
EC2 Stop Schedule Configuration

- **Cron Expression:** cron(59 14 * * ? *)
- **Execution Time:** 2:59 PM IST

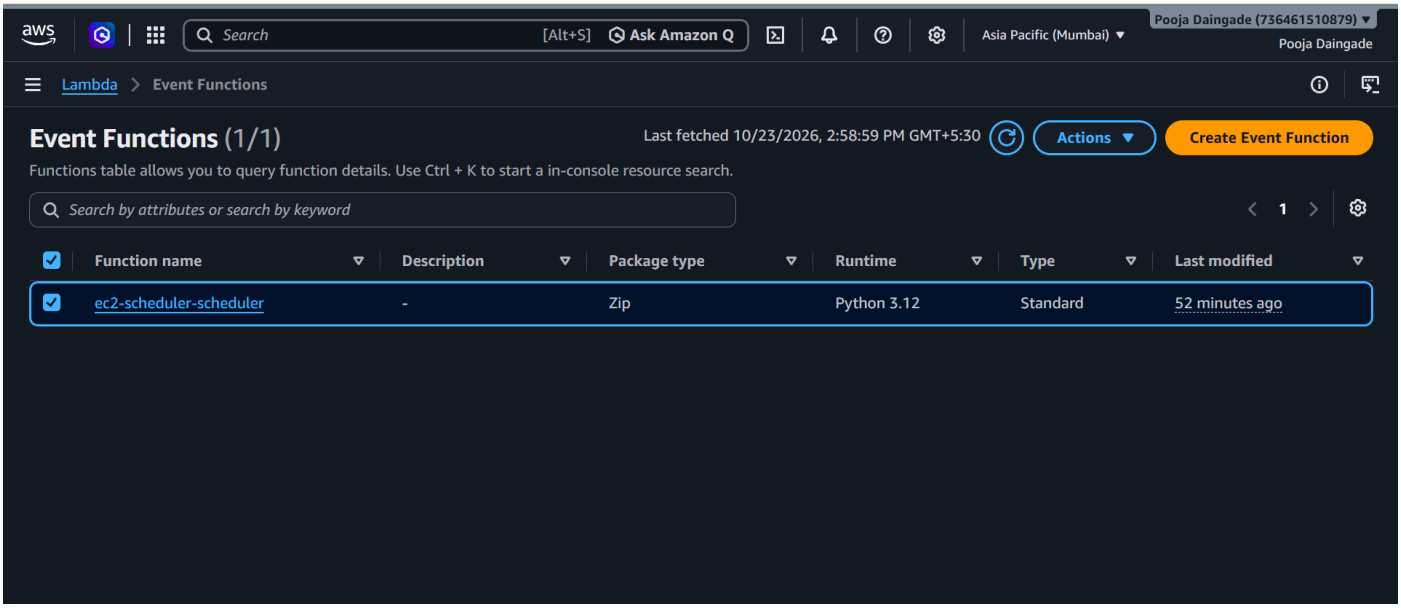


EC2 Start Schedule Configuration

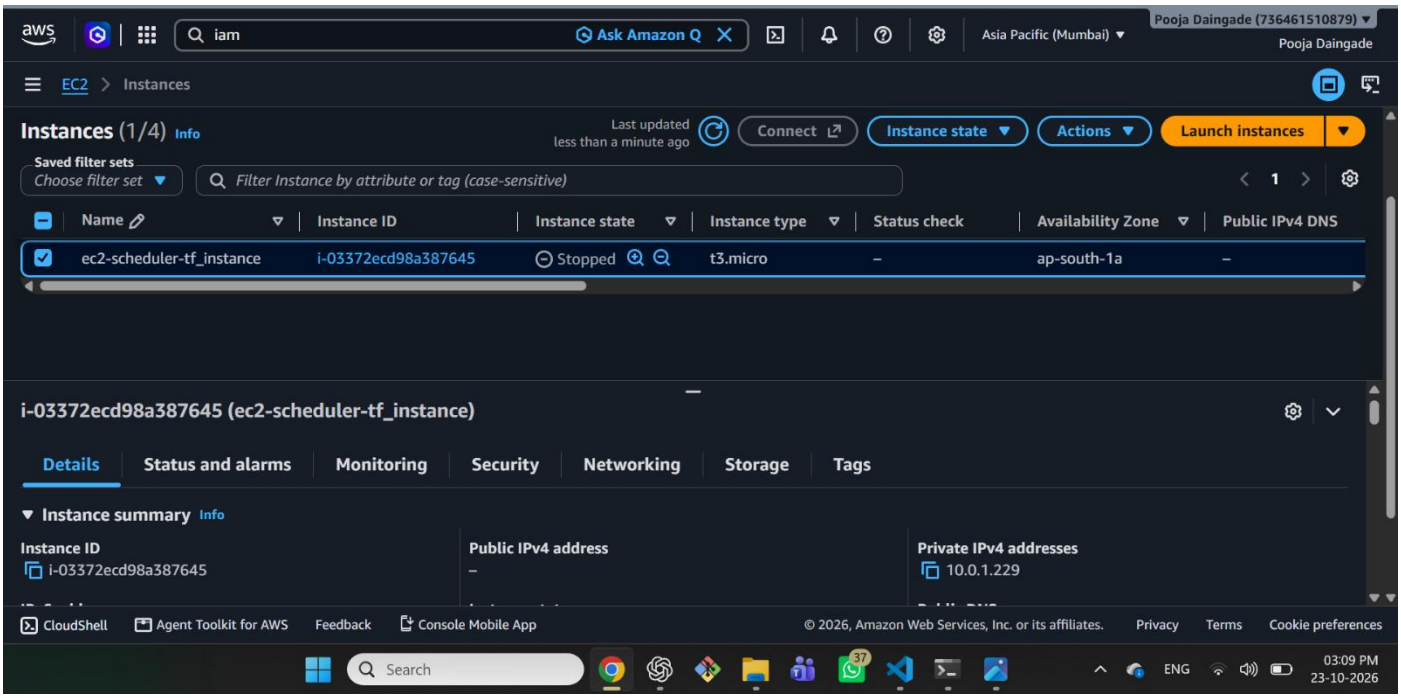
- **Cron Expression:** cron(10 15 * * ? *)
- **Execution Time:** 3:10 PM IST



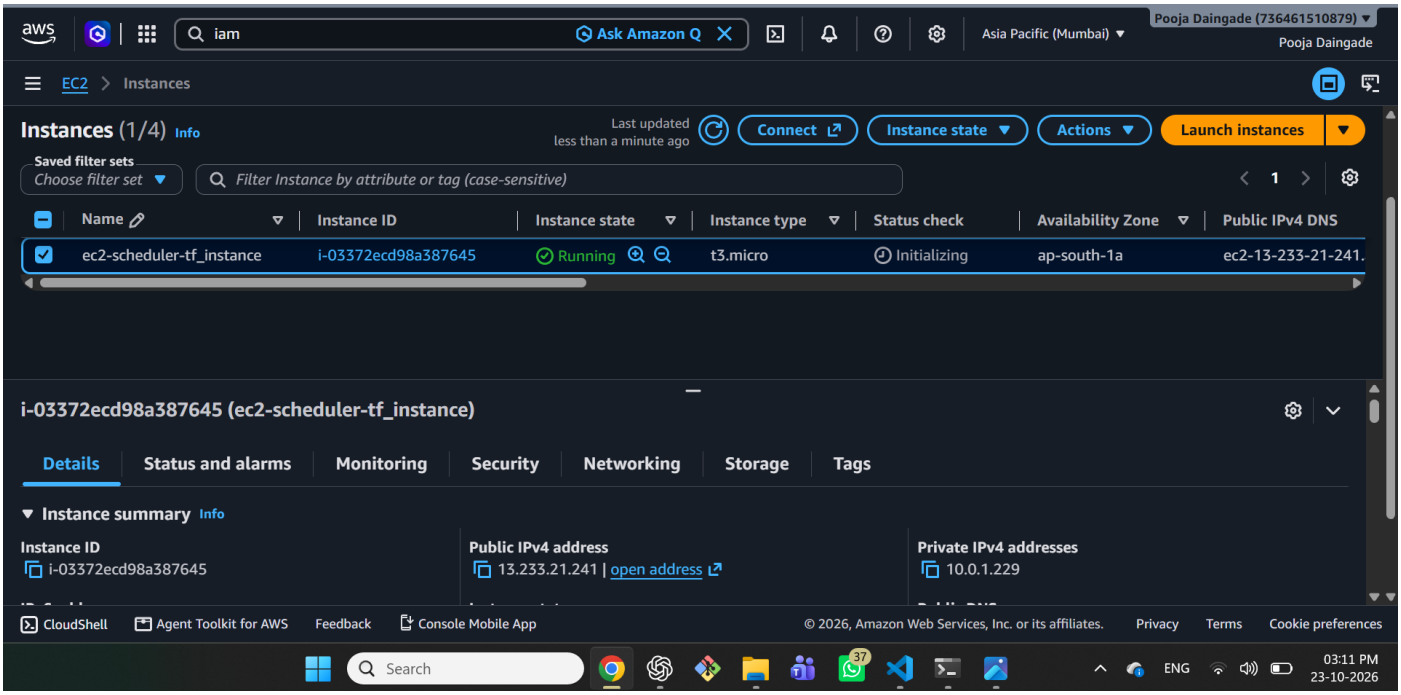
4. AWS Lambda is configured to automate EC2 instance start and stop operations based on EventBridge Scheduler triggers.



5. The EC2 instance state is verified as **Stopped** after the scheduled stop operation. This confirms the instance has been shut down, helping reduce compute costs when it is not required.



6. The EC2 instance successfully transitioned to the **Running** state through the automated scheduling process. This confirms that the AWS Lambda function can initiate EC2 startup operations as configured.



Conclusion

This implementation demonstrates automated EC2 instance start and stop operations using Terraform, AWS Lambda, and Amazon EventBridge Scheduler. It reduces manual intervention, optimizes resource usage, and helps control AWS costs through scheduled instance management with IAM permissions.